

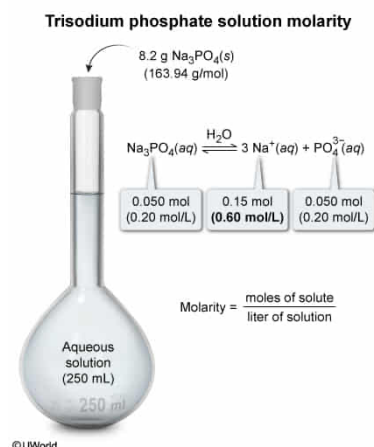
What is the approximate molar concentration of Na^+ ions in a solution prepared by dissolving 8.2 g of Na_3PO_4 (MM = 163.94 g/mol) into 250 mL of water?

- ☐ A. 6.0×10^{-4} M [7%]
☐ B. 3.3×10^{-2} M [15%]
☐ C. 2.0×10^{-1} M [36%]
☒ D. 6.0×10^{-1} M [41%]

Correct

40% answered correctly

Explanation:

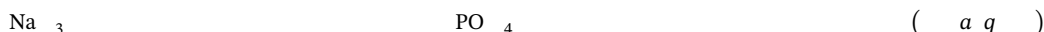


The **molar concentration (molarity)** of a chemical species is defined as the **number of moles** of the species **per liter of solution (mol/L)**. If a concentration is known in terms of the mass of solute per unit volume, the mass must be converted into moles and the volume expressed in liters to correctly determine the molarity.

The mass of a compound sample can be converted into the equivalent number of moles using the molar mass, which is the amount of mass contained in 1 mole of the compound (grams per mole). In this question, dividing the 8.2 g of Na_3PO_4 in the solution by the molar mass of Na_3PO_4 gives:

$$8.2 \text{ g} \div 163.94 \text{ g/mol} = 0.050 \text{ mol}$$

The moles of one chemical species can be related to the moles of another chemical species by the **mole ratios** (stoichiometry) from a chemical reaction or compound formula. The formula of Na_3PO_4 shows that 3 moles of Na^+ ions are released into solution for every 1 mole of Na_3PO_4 that dissolves:



Applying this 3:1 ratio gives the Na^+ ions released into solution:

$$0.050 \text{ mol } \text{Na}_3\text{PO}_4 \times \frac{3 \text{ mol } \text{Na}^+}{1 \text{ mol } \text{Na}_3\text{PO}_4} = 0.15 \text{ mol } \text{Na}^+$$

Dividing the moles of Na^+ ions in solution by the total volume of solution expressed in liters (0.25 L) yields:

$$\frac{0.15 \text{ mol } \text{Na}^+}{0.25 \text{ L}} = 0.60 \text{ mol/L}$$

Therefore, the molar concentration of Na^+ ions is 6.0×10^{-1} M.

(Choice A) A concentration of 6.0×10^{-4} mol/mL is calculated if the solution volume is not converted to liters. Molarity is defined as moles per liter rather than moles per milliliter.

(Choice B) Expressing the concentration in terms of mass per unit volume gives 3.3×10^{-2} g/mL of Na_3PO_4 instead of the molar concentration (mol/L) of Na^+ ions.

(Choice C) A concentration of 2.0×10^{-1} M is the molarity of the Na_3PO_4 rather than the molarity of the Na^+ ions. The 3:1 mole ratio of Na^+ to Na_3PO_4 makes the Na^+ concentration three times higher than that of Na_3PO_4 .

Educational objective:

The molar concentration (molarity) of a chemical species is defined as the number of moles of the species per liter of solution (mol/L).

Time spent: 0 sec
Subject:
General Chemistry

QID: 402751
System:
Atomic Theory and Chemical Composition